HOW TO INSTALL ARM64 GENTOO LINUX ON UTM IN 18 STEPS | MASBOOK FOR ARM64 UTM

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Be prepared to be very patient, as the installation can take hours :)

0.BEFORE START

***PC Keyboard RECOMMENDED**

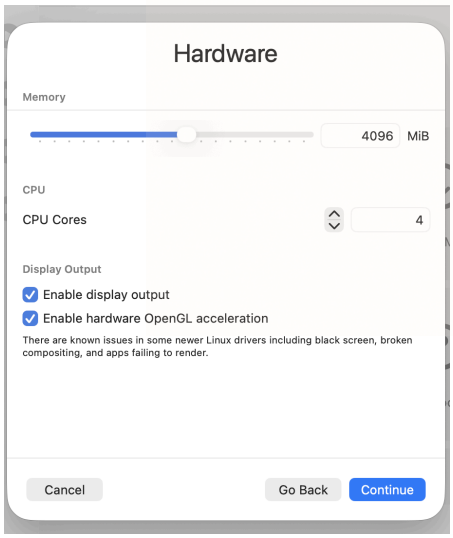
***If you press a key while holding Shift, it becomes uppercase.**

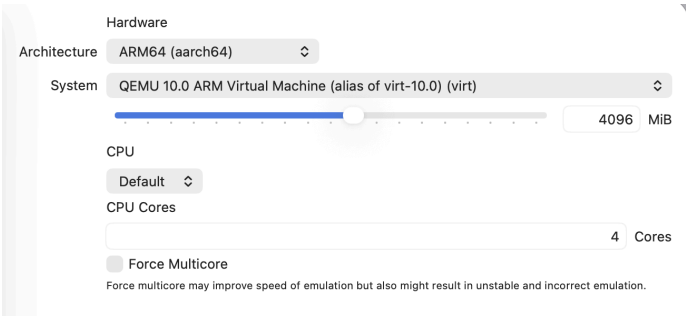
***To type the '|' symbol on a Mac, press "Option" + "-" at the same time. On a standard keyboard, press "AltGr" + the key with the "<,>" symbol.**

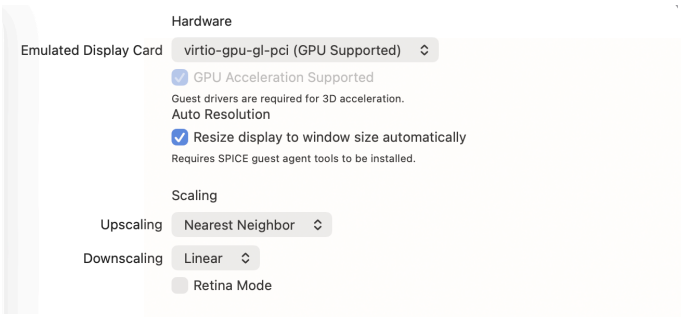
***By pressing "Ctrl" + "C", you can stop an action or cancel a running process.**

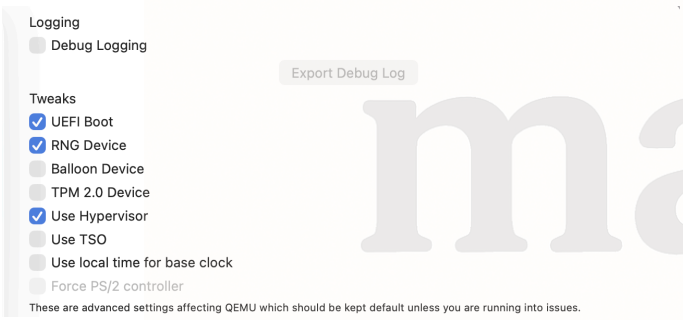
***To exit in many cases, first press "ESC", then type ":wq"**

***Add a space where (space) is indicated; otherwise, the commands must be written without any (space)'s.**









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1. START (SET YOUR KEYBOARD BEFORE START / DEFAULT IS ENGLISH) (trq = Turkish Q)

loadkeys (space)trq

ping (space)-c(space) 3 (space)google.com

2. DISK (USE ARROW KEYS TO NAVIGATE)

cfdisk /dev/vda

GPT

1 GB -> EFI FILE SYSTEM
4 GB -> LINUX SWAP
REMAINING STORAGE -> LINUX FILE SYSTEM
(DON'T FORGET TO WRITE IT AFTER TYPE)

3. FORMAT + MOUNT

mkfs.ext4(space) /dev/vda3

mkfs.fat(space) -F 32(space) /dev/vda1

mkswap(space) /dev/vda2

mkdir(space) -p (space) /mnt/gentoo

mount(space) /dev/vda3(space) /mnt/gentoo

swapon(space) /dev/vda2

4. STAGE3

cd(space) /mnt/gentoo

links(space) https://www.gentoo.org/downloads/

ls

tar(space) xpvf(space) stage3-arm64-openrc-20260315T234601Z.tar.xz(space)--xattrs-include='*.*'(space)--numeric-owner **(TYPE STAGE AND PRESS THE TAB KEY TO AUTO-COMPLETE)**

ls

vi(space) /mnt/gentoo/etc/portage/make.conf

MAKEOPTS=”-j4”

GRUB_PLATFORMS="arm64-efi”

VIDEO_CARDS="virtio”

5. MOUNT SYSTEM

cp(space)--dereference(space) /etc/resolv.conf(space)/mnt/gentoo/etc

mount (space)--types (space)proc(space)/proc(space)/mnt/gentoo/proc

mount (space)--rbind(space)/sys(space)/mnt/gentoo/sys

mount (space)--make-rslave (space)/mnt/gentoo/sys

mount (space)--rbind (space)/dev(space)/mnt/gentoo/dev

mount (space)--make-rslave (space)/mnt/gentoo/dev

mount (space)--bind(space)/run(space)/mnt/gentoo/run

mount (space)--make-slave(space) /mnt/gentoo/run

6. CHROOT

chroot(space) /mnt/gentoo(space)/bin/bash

source (space)/etc/profile

export (space)PS1="(ch) (space) $PS1"

mkdir(space) -p(space)/boot/efi

mount(space) /dev/vda1(space) boot/efi

7. UPDATE

emerge-webrsync

emerge(space)--sync(space)--quiet

emerge(space)--ask(space)--verbose(space)--update(space)--deep(space)--changed-use(space)@world

emerge(space)-q(space)app-editors/vim

8. TIME (for Istanbul, change location to yours /zoneinfo/Country/City)

ln(space)-sf(space)../usr/share/zoneinfo/Turkey/Istanbul(space)/etc/localtime

9. LOCALE

vim(space)/etc/locale.gen

locale-gen

eselect(space)locale(space)list

eselect(space)locale(space)set(space)4

env-update(space)&&(space) source(space) /etc/profile(space) &&(space) export(space) PS1="(ch)(space) $PS1"

10. KERNEL

echo(space) ‘sys-kernel/linux-firmware (space)@BINARY-REDISTRIBUTABLE’(space) | (space) tee (space)-a (space)/etc/portage/package.license

emerge(space)-q(space)sys-kernel/linux-firmware

emerge(space)--ask(space)sys-firmware/sof-firmware

echo(space)”sys-kernel/installkernel(space)dracut(space)grub”(space)>>(space)/etc/portage/package.use/installkernel

emerge(space)-q(space)sys-kernel/installkernel

emerge(space)-q(space)sys-kernel/gentoo-kernel-bin

uname(space)-r

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11. FSTAB

```
# <fs>                  <mountpoint>    <type>          <opts>          <dump> <pass>

#LABEL=boot              /boot           ext4            defaults        1 2
#UUID=58e72203-57d1-4497-81ad-97655bd56494                /               xfs             defaults                0 1
#LABEL=swap              none            swap            sw              0 0
#/dev/cdrom              /mnt/cdrom      auto            noauto,ro       0 0
/dev/vda1                /boot/efi       vfat            defaults        0 2
/dev/vda2                none            swap            sw              0 0
/dev/vda3                /               ext4            noatime         0 1
```

vim(space)/etc/fstab

12. NETWORK

vim(space) /etc/conf.d/hostname

Press J
Press F
Press C

Press I
Press W

vim(space)/etc/hosts

Press SHIFT + V
Press Y
Press P
Press F
Press L
Press C
Press I
Press W

emerge (space)-q(space)net-misc/dhcpcd

emerge(space)-q(space)--noreplace(space)net-misc/netifrc

ifconfig(mine is : enp0s1)

vim(space)/etc/conf.d/net

config_enp0s1="dhcp"

cd(space)/etc/init.d

ln(space)-s(space)net.lo(space)net.enp0s1

rc-update(space)add(space)net.enp0s1(space)default

13. ROOT + USER

passwd

useradd(space)-m(space)-G(space) wheel,audio,video
(space)-s(space)/bin/bash(space) user

passwd (space) user

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14. SUDO

emerge(space)-q(space)sudo

EDITOR=vim(space)visudo

wheel (space)ALL=(ALL: ALL) (space) ALL

15. BOOTLOADER

cd

echo(space)'GRUB_PLATFORMS="arm64-efi"'(space)>>(space)/etc/portage/make.conf

emerge(space)-q(space)sys-boot/grub(space)efibootmgr(space)neofetch

mount(space)-t(space)efivarfs(space)efivarfs(space)/sys/firmware/efi/efivars

vim(space) /etc/default/grub

GRUB_CMDLINE_LINUX_DEFAULT="quiet video=virtio-gpu"

grub-install(space) --target=arm64-efi(space) --efi-directory=/boot/efi

grub-mkconfig(space)-o(space)/boot/grub/grub.cfg

find(space) / (space)-name(space) "vmlinuz*"(space) -o -name(space) "Image"

cp(space) /usr/src/linux-6.18.18-gentoo-dist/arch/ arm64/boot/vmlinuz.efi(space) /boot/vmlinuz-gentoo

emerge(space) -q(space) sys-kernel/dracut

dracut(space) --force(space) /boot/initramfs-gentoo.img(space) 6.18.18-gentoo-dist

grub-mkconfig(space) -o(space) /boot/grub/grub.cfg

uname(space) -r

16. FINISH

exit

cd(space) /

umount (space)-R(space)/mnt/gentoo

reboot

17. AFTER REBOOT

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user

1234

17.1 Keyboard Layout Configuration (trq = Turkish Q , Set Your Language)

sudo(space) nano (space)/etc/conf.d/keymaps

keymap="trq"

sudo(space) rc-service (space)keymaps(space) restart

17.2 Xorg and Essential Tools Installation

sudo(space) touch(space) /etc/portage/package.use/xorg

sudo (space) emerge (space) --ask (space) -q (space) xorg-server (space) x11-apps/xinit (space) x11-apps/xrandr (space) xorg-drivers (space)elogind (space) dev-vcs/git

sudo (space) dispatch-conf (then press u to update)

sudo(space) emerge(space) --ask (space) -q (space) xorg-server (space) x11-apps/xinit (space) x11-apps/xrandr (space)xorg-drivers (space)elogind (space) dev-vcs/git

ping (space) -c (space)3 (space) google.com

17.3 Service Management and Network Setup

sudo (space) emerge (space) --ask (space) -q (space) xorg-server (space) x11-apps/xinit (space) x11-apps/xrandr (space) xorg-drivers (space) elogind (space) dev-vcs/git

sudo(space) rc-update(space) add(space) elogind(space) boot

sudo (space)emerge (space) -q(space) net-misc/networkmanager

sudo (space) rc-update (space) add (space) NetworkManager (space)default

sudo (space) echo (space) "net-wireless/wpa_supplicant(space) dbus" (space) | (space) sudo (space) tee (space) -a (space) /etc/portage/package.use/xorg

sudo (space)dispatch-conf

sudo (space) emerge (space) --ask(space) -q (space) net-misc/networkmanager

sudo (space) rc-update (space) add (space) NetworkManager(space) default

17.4 Window Manager (DWM) and Terminal (ST)

sudo (space) rc-update(space)add (space) elogind (space) boot

sudo (space) emerge (space) -q (space)dwm (space) st

echo (space) "exec(space) dwm" (space) > (space) .xinitrc

17.5 Launching the Environment and Verification

sudo (space) reboot

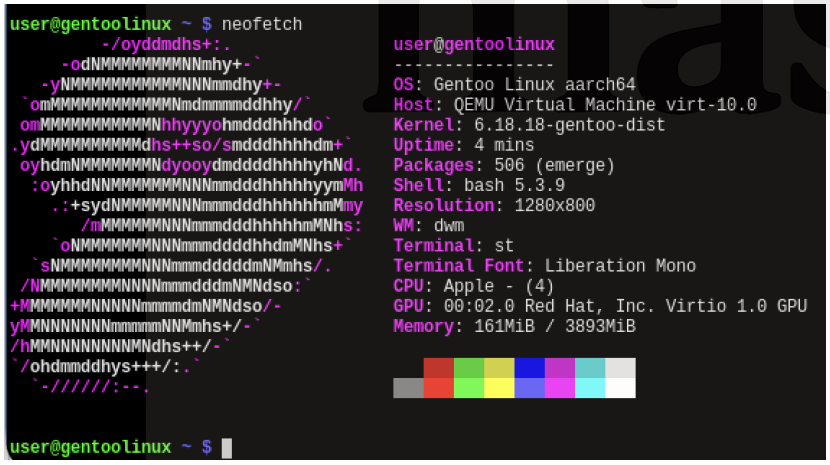
startx

Option + Shift + Enter

ping(space) -c(space) 4(space) google.com

sudo (space) emerge (space) -q(space) app-misc/neofetch

neofetch



18. KDE PLASMA INSTALLATION

NOTE: Before starting this process, shut down the system by typing sudo(space)poweroff in the terminal. Then, restart the

virtual machine. After logging into your user account, proceed with the steps below.

18.1 Environment and Profile Setup

sudo(space) su(space)-

nano(space)/etc/portage/make.conf

USE="kde(space)plasma(space)qt6(space)wayland(space)dbus(space) sddm(space)-gnome(space) -gtk"

eselect(space)profile(space)list

eselect(space)profile(space)set(space)9

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18.2 Network and Dependency Prep

ping(space)-c(space)3(space)8.8.8.8

rc-service(space)NetworkManager(space)restart

USE="-tiff"(space)emerge(space) --ask(space)--oneshot(space)media-libs/libwebp

18.3 System Update and Core Installation

emerge(space)--ask(space)--verbose(space)--update(space)--deep(space)--newuse(space)@world

etc-update

-3

y

emerge(space)--ask(space)kde-plasma/plasma-meta

emerge(space)--ask(space)x11-misc/sddm

18.4 Service and Display Management

rc-update(space)add(space)dbus(space)default

rc-update(space)add(space)sddm(space)default

rc-update(space)add(space)NetworkManager(space)default

ls(space)/etc/init.d/

echo(space)'DISPLAYMANAGER="sddm"'(space)>(space)/etc/conf.d/display-manager

rc-update(space)add(space)display-manager(space)default

reboot

18.5 Display and Peripheral Optimization

***Display Configuration:** Set resolution to 1920x1200 and scaling to 120%.
***Keyboard Layout:** Set to Apple | Apple Aluminium (ISO)

emerge(space)--ask(space)app-emulation/spice-vdagent

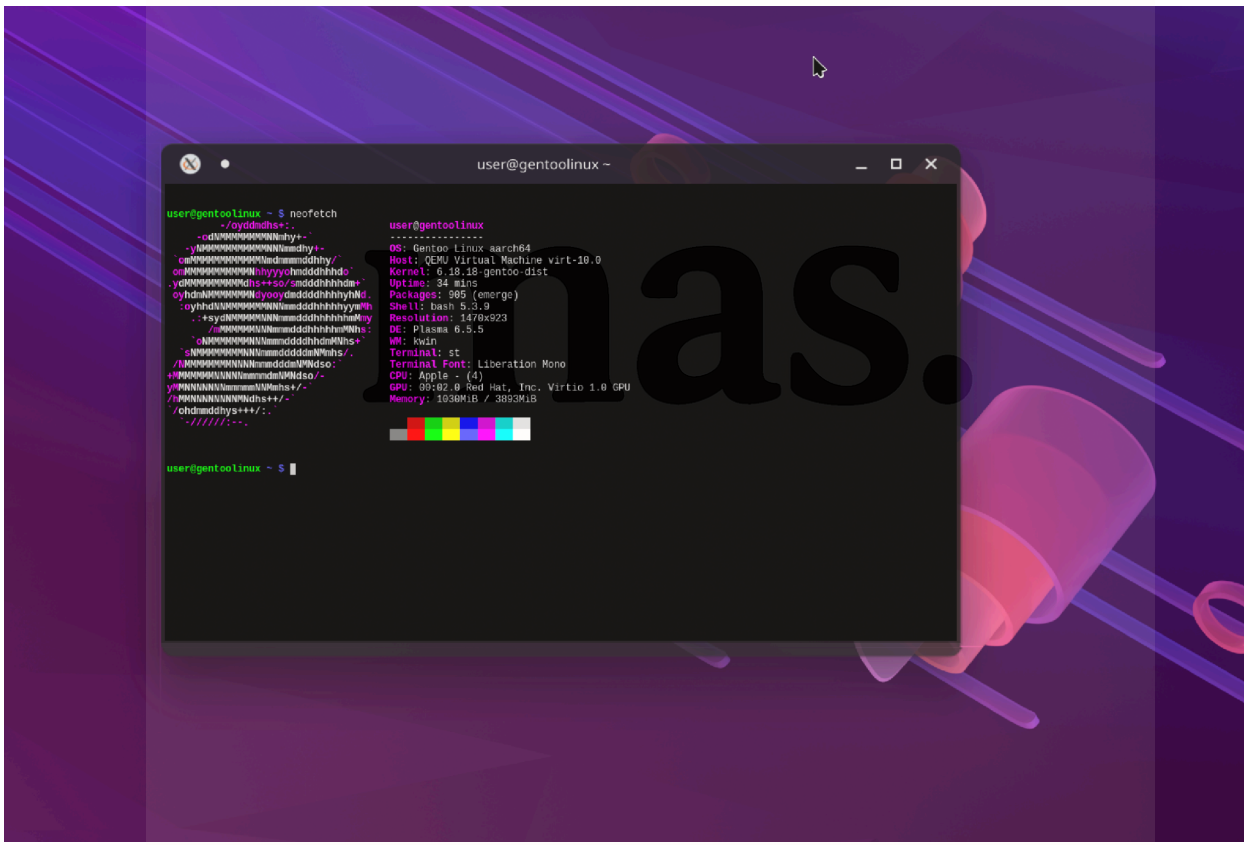
sudo(space)mkdir(space)-p(space)/etc/portage/package.accept_keywords

echo(space)"app-emulation/spice-vdagent(space)~arm64"(space)|(space)sudo(space)tee(space)-a(space)/etc/portage/package.accept_keywords/spice

sudo(space)emerge(space)--ask(space)app-emulation/spice-vdagent

sudo(space)rc-update(space)add(space)spice-vdagent(space)default

sudo(space)rc-service(space)spice-vdagent(space)start

18.6 Web Browser Installation

sudo(space)emerge(space)--ask(space)www-client/firefox-bin